

Hydroecological pressure signals in Mingachevir Reservoir: A retrospective synthesis of nutrient, microbial and major-ion evidence from the Middle Kura cascade

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Keywords

Mingachevir Reservoir; Kura River; water quality; microbial abundance; nutrients; mineralization; reservoir monitoring

ABSTRACT

Mingachevir Reservoir is the principal regulating water body of Azerbaijan's Middle Kura cascade, yet much of its hydroecological evidence remains dispersed across legacy monitoring tables and local studies. This paper consolidates station-level microbial observations, nutrient records, spatial hydrochemical measurements and major-ion profiles reported in a 2023 master's dissertation, recalculates internally inconsistent summary values from the underlying station data, and interprets the historical baseline against recent Kura River research published in 2025. Seasonal mean microbial abundance was higher in 2015 than in 1982 in all four seasons, with the largest relative difference in winter (68%). Nutrient records for 1987–2009 indicated pronounced enrichment episodes, particularly for nitrite and phosphorus, although differences in historical analytical protocols require cautious interpretation. Along the Shikhli–Salyan transect, the mean sum of major ions declined through the reservoir cascade, reaching its lowest value at Mingachevir (approximately 560 mg L⁻¹), before increasing downstream to approximately 1,066 mg L⁻¹ at Salyan. Within-reservoir contrasts showed lower transparency and higher mineralization, chloride and iron near the dam zone. Recent studies confirm that mineralization and microbial contamination remain relevant concerns in the Kura system. The combined evidence supports a risk-based monitoring design integrating seasonal field sampling, microbial indicators, nutrients, major ions and satellite-supported observations. The paper provides a transparent historical benchmark rather than claiming contemporary field status.

INTRODUCTION

Large reservoirs are hybrid socio-ecological infrastructures. They buffer seasonal discharge, generate electricity, supply irrigation networks and reduce flood peaks, but they also interrupt sediment transport, alter residence times, reorganize thermal and oxygen regimes, and retain or transform nutrients and contaminants. These processes are especially important in semi-arid basins where flow regulation and consumptive use interact with climate variability. The Kura–Aras basin is a prominent example because upstream pressures are transmitted across borders and accumulate toward lowland Azerbaijan (Mamedov et al., 2005; IPCC, 2023).

Mingachevir Reservoir, impounded on the Kura River in the early 1950s, is the largest reservoir in the South Caucasus and the central storage component of Azerbaijan's water–energy–food system. Its functions include hydropower production, regulation of the Middle and Lower Kura, irrigation supply through the Upper Karabakh and Upper Shirvan canals, fisheries and recreation. This multifunctionality raises the cost of ecological deterioration: changes in water quality affect not only aquatic biota but also downstream agriculture, potable-water treatment and infrastructure operation.

Earlier studies described the reservoir's microbiological regime, nutrient dynamics, organic pollution and major-ion composition, but

the evidence is fragmented and often presented in tables without integrated interpretation (Salmanov & Ansarova, 2016a, 2016b). Some historical series contain inconsistent decimal notation or summary rows that do not reproduce the reported station observations. At the same time, recent research along the Kura has renewed attention to major-ion trends and faecal-indicator organisms, demonstrating the continuing relevance of mineralization and microbial safety in 2025 (Abdyyev et al., 2025; Aliyev & Abdyyev, 2025). A transparent synthesis is therefore useful both for preserving the historical baseline and for designing contemporary monitoring.

This study asks three questions: (1) what seasonal and temporal pressure signals are visible in the legacy microbial and nutrient records; (2) how does major-ion concentration vary along the Middle–Lower Kura cascade; and (3) which monitoring priorities follow when the historical evidence is evaluated alongside recent literature? The objective is not to label the present-day reservoir condition from old measurements. Instead, the study reconstructs an auditable baseline, identifies robust patterns and uncertainties, and proposes a monitoring architecture suitable for current reservoir management.

LITERATURE REVIEW

Reservoir water quality is governed by the interaction of catchment inputs, hydraulic residence time, stratification, sediment exchange and

operational drawdown (Thornton et al., 1990; Wetzel, 2001). Damming can reduce suspended loads and temporarily retain nutrients, while longer residence time may favour algal growth and oxygen depletion. The direction of change is therefore parameter-specific: a reservoir may attenuate downstream mineral loads while simultaneously accumulating organic matter or supporting microbial growth. Global river research also shows that flow regulation interacts with pollution, habitat fragmentation and consumptive use, so multi-parameter assessment is preferable to reliance on a single index (Vörösmarty et al., 2010; Grill et al., 2019).

For the Kura–Aras basin, water-quality pressures arise from municipal wastewater, agriculture, industry, mining and altered flow regimes. The transboundary character of the system complicates source attribution and requires comparable monitoring protocols (Mamedov et al., 2005). Historical Azerbaijani research documented substantial changes in the physical, chemical and microbiological characteristics of Middle Kura reservoirs and highlighted the role of seasonality, pollution inputs and internal sediment processes (Mammadov, 2004; Salmanov & Ansarova, 2016b).

Two recent studies materially update this context. Aliyev and Abduev (2025) analysed major ions at 14 Kura River sites and reported spatial variability in mineralization, including the influence of downstream pollution and hydrogeochemical inputs. Abduev et al. (2025) assessed total coliforms, *Escherichia coli*, enterococci and total viable counts, showing that microbiological surveillance remains necessary along the river. These studies do not replace reservoir-specific sampling, but they confirm that the pressure categories present in the historical Mingachevir record remain management-relevant.

Climate change adds a further layer of uncertainty. Higher temperatures intensify evaporation and can strengthen stratification, while shifts in precipitation and snowmelt affect inflow volume, dilution and pollutant transport (IPCC, 2023). In regulated basins, ecological interpretation should therefore combine water chemistry and microbiology with hydrology, meteorology and remote sensing. Such integration is consistent with contemporary risk-based monitoring, which allocates higher sampling frequency to periods and locations with a greater probability of threshold exceedance.

MATERIALS AND METHODS

Study area and evidence base

Mingachevir Reservoir lies on the Kura River in north-western Azerbaijan, upstream of Mingachevir city and downstream of the Shamkir and Yenikend reservoirs. The reservoir regulates flow toward the Lower Kura and supplies major irrigation canals. The analysed evidence was extracted from the master's dissertation "Investigation of the ecological indicators of Mingachevir Reservoir" (Arifzade, 2023). Four internally coherent datasets were selected: eight-station seasonal microbial observations for 1982 and 2015; nutrient concentrations for 1987–2009; three within-reservoir hydrochemical observations; and three-observation major-ion profiles at six locations from Shikhli to Salyan. Tables lacking numerical content or adequate variable definitions were excluded.

Data harmonization and analysis

Decimal commas were converted to decimal points, units were retained as reported, and station-level values were transcribed without interpolation. For the microbial dataset, seasonal means were recalculated from the eight station values because the reported summer 1982 summary (12.3) was inconsistent with the underlying observations; the reproducible mean is 18.375. Relative difference was calculated as $RD = ((M_{2015} - M_{1982})/M_{1982}) \times 100$. Because the

original microbial unit was not consistently defined, results are reported as "legacy units" and interpreted comparatively rather than against health standards.

The nutrient series was plotted on a logarithmic axis because values span several orders of magnitude. No formal trend test was applied: the series contains only six irregularly spaced years and may combine changing analytical procedures. For the major-ion profile, the arithmetic mean and observed range were calculated from three values at each location. The within-reservoir matrix included transparency, oxygen saturation, mineralization, chloride and iron. Each variable was standardized as $z = (x - \text{mean})/\text{standard deviation}$ solely to visualize relative spatial contrast; raw values remain printed within the heat map.

Interpretation followed a weight-of-evidence approach. A pattern was considered comparatively robust when it was reproduced across stations or aligned with recent peer-reviewed Kura studies. Isolated extreme values were treated as signals requiring verification, not as definitive exceedances. All calculations and figures were produced from the dissertation tables; no values were inferred from photographs or charts.

RESULTS AND DISCUSSION

Seasonal microbial signal

Recalculated seasonal means were higher in 2015 than in 1982 across all seasons (Figure 1). Winter increased from 7.1 to 12.0 legacy units (68.4%), spring from 13.6 to 15.2 (11.9%), summer from 18.4 to 21.9 (19.0%) and autumn from 17.5 to 20.9 (19.3%). The warm-season maximum in both years is consistent with temperature-sensitive microbial activity and longer effective residence time. The large relative winter difference is also important because it suggests that contamination and biological activity cannot be assumed to disappear outside the recreational season.

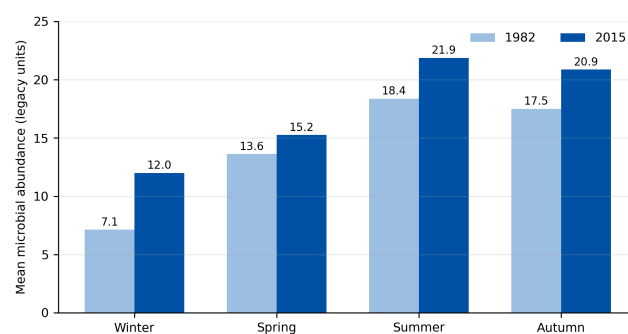


Figure 1. Recalculated seasonal mean microbial abundance at eight Mingachevir Reservoir stations, 1982 and 2015. Source: author's calculations from Arifzade (2023).

The comparison should not be interpreted as a statistically controlled 33-year trend because the record contains only two cross-sections and lacks uncertainty estimates. Nevertheless, the station-consistent increase provides a defensible historical pressure signal. Recent Kura sampling based on total coliforms, *E. coli*, enterococci and viable counts similarly identifies microbiological contamination as an active management concern (Abduev et al., 2025). A modern programme should replace undefined aggregate counts with method-specific indicators, detection limits and quality-control blanks.

Nutrient enrichment episodes

The 1987–2009 nutrient record shows substantial temporal variability (Figure 2). Ammonium-N increased from 0.003 mg L⁻¹ in 1987 to 0.25 mg L⁻¹ in 2007 before declining to 0.08 mg L⁻¹ in 2009. Reported nitrite-N rose sharply and remained elevated after 1993, reaching 3.59 mg L⁻¹ in 2007. Phosphorus peaked at 0.80 mg

L-1 in the same year, whereas nitrate-N reached its highest recorded value (0.13 mg L⁻¹) in 2004. The co-occurrence of ammonium and phosphorus maxima in 2007 is compatible with an episodic loading signal, but the magnitude of the nitrite values also raises the possibility of differing laboratory methods, units or transcription conventions.

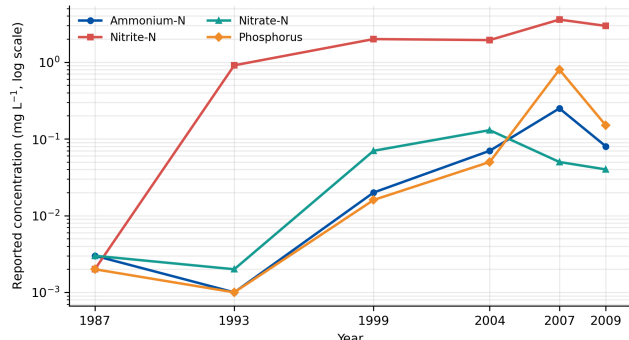


Figure 2. Reported nutrient concentrations in Mingachevir Reservoir, 1987–2009. A logarithmic axis is used because the legacy values span several orders of magnitude. Source: Arifzade (2023).

These uncertainties are scientifically consequential. Without metadata on sampling depth, preservation, analytical method and detection limit, old nutrient values should not be compared mechanically with current regulatory thresholds. Their value is instead diagnostic: they identify nitrogen species and phosphorus as parameters that require renewed, quality-assured sampling. Monthly sampling should be intensified during spring inflow, summer stratification and autumn turnover, with paired surface and near-bottom samples at deep stations.

Longitudinal major-ion pattern

The major-ion sum displayed a clear longitudinal structure (Figure 3). The mean fell from approximately 984 mg L⁻¹ at the Kura–Shikhli station to 655 mg L⁻¹ in Shamkir, 586 mg L⁻¹ in Yenikend and 560 mg L⁻¹ in Mingachevir Reservoir. It then increased to 799 mg L⁻¹ at Zardab and 1,066 mg L⁻¹ at Salyan. The observed range was widest at Shikhli and Salyan, while Mingachevir showed the narrowest range among the six locations. This pattern is consistent with dilution, mixing and retention within the reservoir cascade followed by downstream accumulation from saline groundwater, return flows, evaporation and anthropogenic inputs.

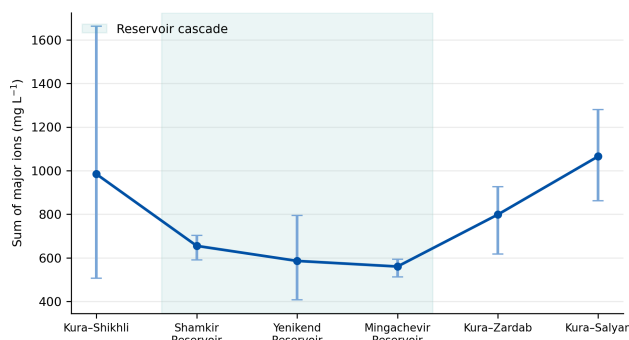


Figure 3. Mean and observed range of the sum of major ions along the Kura–reservoir cascade. Each location contains three legacy observations. Source: author's calculations from Arifzade (2023).

The recent 14-site analysis of Aliyev and Abduev (2025) strengthens this interpretation by documenting contemporary spatial variability in Kura River major ions and increasing mineralization in downstream reaches. The agreement is notable because the datasets differ in date and sampling design. It supports the use of specific conductance and major ions as low-cost sentinel variables, while isotope or

mixing-model work would be needed to separate geological, irrigation-return and wastewater sources.

Table 1. Sum of major ions along the Middle–Lower Kura profile (mg L⁻¹).

Location	Mean	Minimum	Maximum
Kura–Shikhli	984	506	1,662
Shamkir Reservoir	655	590	703
Yenikend Reservoir	586	408	795
Mingachevir Reservoir	560	512	593
Kura–Zardab	799	618	926
Kura–Salyan	1,066	862	1,280

Within-reservoir spatial contrasts

The three-zone snapshot reveals a coherent gradient for several parameters (Figure 4). Transparency decreased from 1.8 near the city-adjacent station to 1.0 in the dense-vegetation zone and 0.5 near the dam. Over the same sequence, mineralization increased from 290 to 450 mg L⁻¹, chloride from 14 to 37 mg L⁻¹ and iron from 0.04 to 0.08 mg L⁻¹. Oxygen saturation was lowest in the dense-vegetation zone (65%) and highest near the dam (92%), illustrating that chemical concentration and oxygen status need not vary in parallel.

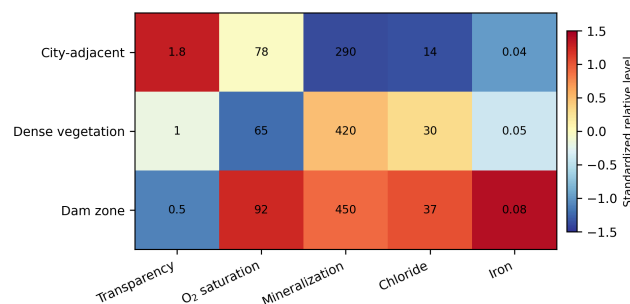


Figure 4. Standardized relative levels of selected hydrochemical indicators across three reservoir zones; cell labels show the original reported values. Source: Arifzade (2023).

Because each zone is represented by one legacy observation, the gradient cannot establish a permanent spatial ranking. It nevertheless provides a practical basis for stratified sampling: an inflow or city-adjacent station, vegetated littoral stations, a deep central station, and dam forebay surface and bottom stations. Sampling only at the dam would miss the lower oxygen signal in vegetated water and would not distinguish local from catchment-scale pressure.

Table 2. Minimum monitoring design proposed for Mingachevir Reservoir.

Component	Core variables	Recommended frequency
Hydrology	Level, inflow, outflow, residence time	Continuous / daily
Field profile	Temperature, pH, EC, DO, turbidity	Monthly; depth-resolved
Nutrients	NH ₄ -N, NO ₂ -N, NO ₃ -N, TN, PO ₄ -P, TP	Monthly; biweekly in summer
Microbiology	E. coli, enterococci, total coliforms	Monthly; event-based
Major ions	HCO ₃ ⁻ , SO ₄ ²⁻ , Cl ⁻ , Ca ²⁺ , Mg ²⁺ , Na+K	Quarterly
Remote sensing	Water extent, turbidity proxy, chlorophyll proxy	Monthly cloud-free scenes

Management implications and limitations

The synthesis points to three management priorities. First, monitoring should be seasonal and depth-resolved. The microbial

maximum in warm seasons and the potential for internal loading during stratification or turnover make annual grab sampling inadequate. Second, reservoir observations should be interpreted within the cascade. The decline and subsequent downstream rise in major ions show that an apparent improvement within Mingachevir does not guarantee lower exposure farther downstream. Third, data governance is as important as parameter coverage. Every observation should include coordinates, depth, date and time, hydrological state, method, detection limit, laboratory quality-control information and a stable data identifier.

The principal limitation is the age and heterogeneity of the source data. The study cannot determine the reservoir's current compliance status, estimate causal effects of dam operation, or distinguish changes in environmental conditions from changes in analytical technique. Some original units and sampling protocols were incompletely documented. Recalculation corrects an arithmetic inconsistency but cannot repair missing metadata. The visualized patterns should therefore guide contemporary sampling and hypothesis formation rather than substitute for new field evidence.

A stronger next phase would combine at least two full hydrological years of standardized field measurements with Sentinel-2 or Landsat observations and operational water-level data. Relationships between reflectance-based turbidity or chlorophyll proxies and laboratory measurements should be locally calibrated. Event-based samples after high inflow, wastewater incidents or rapid drawdown would improve attribution. Such a system would connect ecological protection with hydropower, irrigation reliability and treatment-cost reduction, aligning reservoir management with green-economy objectives.

CONCLUSIONS

This study converted dispersed dissertation tables into an integrated hydroecological baseline for Mingachevir Reservoir and the Middle–Lower Kura cascade. Four findings are central. Microbial abundance was higher in 2015 than in 1982 in every season, with the largest relative difference in winter. The nutrient series contained marked enrichment episodes, especially for nitrite and phosphorus, but methodological uncertainty prevents direct regulatory interpretation. The sum of major ions declined through the cascade to a mean of about 560 mg L⁻¹ at Mingachevir and then rose downstream to about 1,066 mg L⁻¹ at Salyan. Within the reservoir, lower transparency and higher mineralization, chloride and iron were reported toward the dam zone, while the lowest oxygen saturation occurred in dense vegetation.

Recent 2025 studies of Kura River ions and microbial indicators show that these historical pressure categories remain relevant. The appropriate response is not to treat legacy values as contemporary status, but to use them to design a reproducible monitoring network. A minimum system should integrate continuous hydrology, depth-resolved field profiles, nutrients, method-specific microbial indicators, quarterly major-ion analysis and calibrated satellite observations. This combination would allow managers to detect seasonal risk, separate reservoir retention from downstream re-enrichment and target interventions where environmental and economic benefits are greatest.

AUTHOR CONTRIBUTIONS

The author was responsible for conceptualization, data curation, formal analysis, visualization, interpretation, writing—original draft, and writing—review and editing.

CONFLICTS OF INTEREST

The author declares no conflict of interest.

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This research received no external funding.

DATA AVAILABILITY

All numerical data analysed in this article are reproduced in the tables and figures or are available in the underlying master's dissertation cited as Arifzade (2023). No new field dataset was generated.

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